Italian Bond Yield Regimes: A Hidden Markov Model

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1 Introduction

This analysis examines changes in Italian government bond yields using a three-state Gaussian Hidden Markov Model. The model allows the mean and variance of nominal yields to depend on an unobserved state, with transitions governed by a Markov chain.

We estimate the model using monthly observations from January 1997 to July 2019 and decode the most likely state sequence. States are labelled by their estimated mean yields rather than interpreted automatically as economic expansions or crises.

2 Data and Exploratory Analysis

The dataset contains 271 monthly observations on nominal 10-year Italian government bond yields, inflation, and additional macroeconomic and credit-rating variables. The HMM uses nominal yields only.

For descriptive comparison, we also display nominal yields minus contemporaneous inflation. This is an approximate inflation-adjusted measure, rather than an ex ante real yield based on expected inflation.

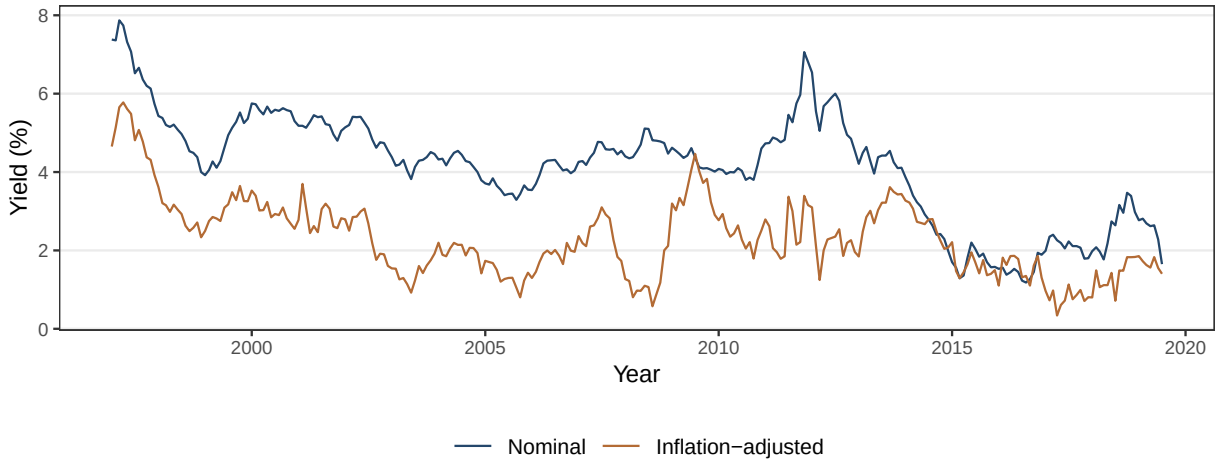


Figure 1. *Monthly Italian 10-year government bond yields, January 1997–July 2019. The inflation-adjusted series subtracts contemporaneous inflation from nominal yields.*

Nominal yields display sustained periods at different levels, including elevated observations near the beginning of the sample and around 2011–2012, followed by lower yields in the later years. These patterns motivate a model with persistent latent states, although the plot alone does not establish that three distinct regimes generate the data.

3 Three-State Gaussian HMM

Let Y_t denote the nominal yield and S_t an unobserved state taking values in $\{1, 2, 3\}$. Conditional on the state sequence, observations are independent, with

$$Y_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2).$$

The latent process follows a first-order Markov chain with time-invariant transition probabilities

$$p_{ij} = \Pr(S_t = j \mid S_{t-1} = i), \qquad \sum_{j=1}^{3} p_{ij} = 1.$$

Each state has its own mean and variance. The model contains no explanatory variables and is estimated on the full sample using the expectation-maximization algorithm. This is a retrospective classification analysis, without a separate forecasting test sample.

State labels are arbitrary during estimation. For presentation, we order states by increasing estimated mean yield and refer to them as low-, medium-, and high-yield states. Numerical estimation can converge to a local likelihood maximum.

4 Parameter Estimates and Uncertainty

States are ordered by increasing estimated nominal yield. The standard deviations below describe within-state dispersion; they are not standard errors of the parameter estimates.

Table 1: Estimated Gaussian emission parameters.

State	Mean yield (%)	SD (percentage points)
Low-yield	2.137	0.581
Medium-yield	4.251	0.377
High-yield	5.639	0.713

The transition matrix reports the probability of moving from the state in each row to the state in each column in one month.

Table 2: Estimated monthly transition probabilities.

	Low-yield	Medium-yield	High-yield
Low-yield	1.000	0.000	0.000
Medium-yield	0.008	0.976	0.016
High-yield	0.000	0.041	0.959

Table 3: Estimated initial-state probabilities.

State	Probability
Low-yield	0
Medium-yield	0
High-yield	1

Estimated mean yields are 2.137%, 4.251%, and 5.639% across the three states. Within-state dispersion is greatest in the high-yield state and smallest in the medium-yield state.

The diagonal transition probabilities indicate high persistence under the fitted model. However, the decoded sequence contains few state changes, and the low-yield state appears in a single spell extending to the end of the sample, with no observed exit in the decoded path. This limited transition information helps explain why its estimated self-transition probability rounds to 1.000; it does not establish a permanently absorbing economic regime.

The estimated initial distribution places essentially all probability on the high-yield state. With only one observed time series, information about the initial distribution is limited. This estimate describes the beginning of this sequence and should not be interpreted as the long-run distribution of states. Approximate standard errors are computed from the numerical observed-information matrix, accounting for the model constraints. Parameters on or near their boundaries are excluded from this calculation. Reported uncertainty is therefore conditional on those boundary parameters.

Table 4: Parameter estimates and approximate standard errors.

Parameter	Estimate	SE
Initial: Low-yield	0.0000	NA
Initial: Medium-yield	0.0000	NA
Initial: High-yield	1.0000	NA
Low-yield to Low-yield	1.0000	NA
Low-yield to Medium-yield	0.0000	NA
Low-yield to High-yield	0.0000	NA
Medium-yield to Low-yield	0.0076	0.0076
Medium-yield to Medium-yield	0.9762	0.0137
Medium-yield to High-yield	0.0162	0.0114
High-yield to Low-yield	0.0000	NA
High-yield to Medium-yield	0.0413	0.0233
High-yield to High-yield	0.9587	0.0233
Low-yield mean	2.1371	0.0740

Parameter	Estimate	SE
Low-yield SD	0.5810	0.0528
Medium-yield mean	4.2509	0.0376
Medium-yield SD	0.3766	0.0257
High-yield mean	5.6390	0.0923
High-yield SD	0.7134	0.0586

NA denotes an unavailable standard error, not a zero standard error. Boundary estimates limit the usual normal approximation and require cautious interpretation. Standard errors for the state means are 0.0740, 0.0376, and 0.0923 percentage points for the low-, medium-, and high-yield states, respectively. Several transition probabilities have substantial uncertainty relative to their estimates, so individual transition patterns should be interpreted cautiously.

5 State Decoding

The Viterbi algorithm identifies the most probable joint state sequence conditional on all observations and the fitted parameters. This is a retrospective classification, rather than a real-time signal.

We compare observed yields with the estimated mean associated with each decoded state.

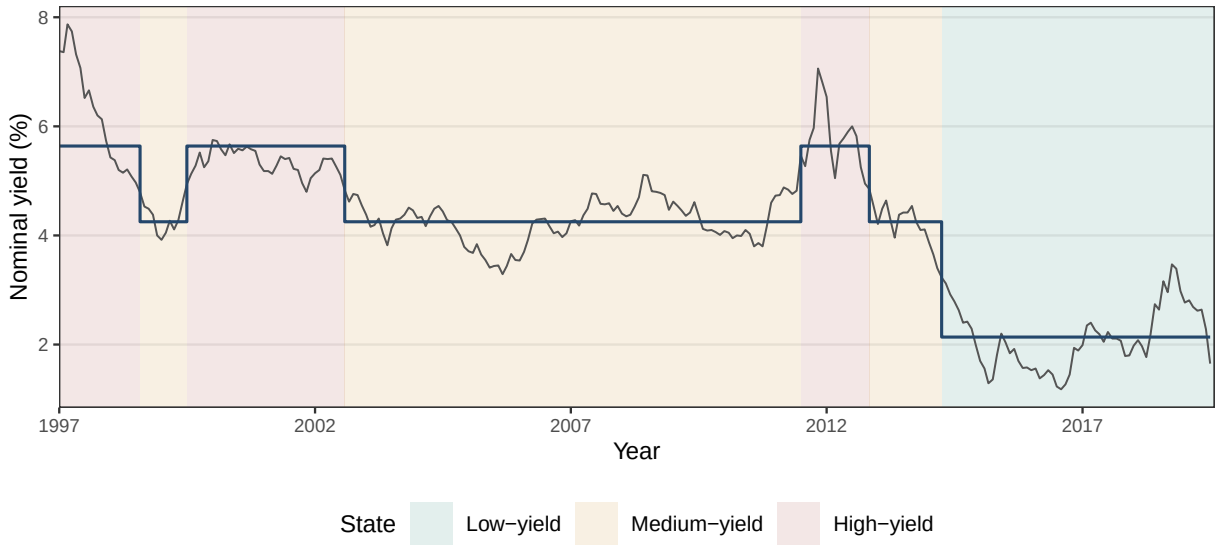


Figure 2. *Observed nominal yields and estimated means along the Viterbi-decoded state sequence. Shading identifies low-, medium-, and high-yield states. The step line represents decoded state means, not one-step-ahead forecasts.*

The high-yield state appears near the beginning of the sample, around 2000–2002, and during 2011–2012. The medium-yield state covers much of the intervening period, while the low-yield state dominates from approximately 2014 onward.

The decoded means summarize persistent level differences while leaving substantial within-state fluctuations. These assignments depend on the fitted model and do not independently identify economic crises or structural breaks.

6 Discussion and Limitations

The three-state Gaussian HMM summarizes nominal Italian bond yields through states with different means, dispersions, and high estimated persistence. The decoded sequence distinguishes sustained periods of low, medium, and high yields.

The number of states is fixed at three, and transition probabilities are assumed constant over time. The model assumes conditional independence of observations given the state sequence; persistent within-state dynamics could limit its adequacy and the reliability of model-based standard errors. Estimation uses a single initialization, so a global likelihood maximum is not established. Boundary probability estimates also limit standard-error inference.

The analysis uses the full sample for estimation and decoding. It therefore provides a retrospective description, without establishing real-time classification or forecasting performance.